# SQCD-1 · Settlement Quality Compute Data

The record a delivered compute hour settles on.

One record covers one delivery claim over one window.

Issued by PSYCHE · Version 1.0 · September 2026

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## 0 · Status and adoption

SQCD-1 is a record standard. It is adopted by reference: two parties name it in their agreement and it governs the form of the record that settles delivery between them.

No regulator administers it. No body approves a filing under it. Every consequence in this standard is evidentiary — it changes what a record proves, not what a party owes. What a party owes is in the contract.

The words MUST, MUST NOT, SHOULD and MAY carry their ordinary drafting force. A requirement marked MUST is a condition of conformance.

**Drafting status. SQCD-1 does not set a value for any term that allocates loss between the parties.** A term whose value decides who pays when the record is silent is a commercial term; a value chosen here would apply uniformly to every contract processed under this standard, whatever any of them say, and would sit where neither party's counsel reads it.

Where a party adopts SQCD-1 with modifications, it MUST declare each modification in its Filing Plan. A modification that is not declared makes the filing non-conforming.

**0.1 · Adopting it by reference.** A contract incorporates this standard by naming the version and the digest of the text:

> *This Agreement incorporates SQCD-1, Version 1.0, identified by the digest published with it, as issued by PSYCHE, and no later version.*

**A reference that names no version names nothing this standard will defend.** Conformance under L5 is to a named version permanently, and a reference to "SQCD-1" without one leaves the parties to argue later about which text they agreed to. That is the argument this standard exists to remove, moved up one level.

**0.2 · Rights in this text.** The text of this standard is published under the **Creative Commons Attribution-NoDerivatives 4.0 International licence**: it may be copied, redistributed and attached to an agreement in full, by anyone, for any purpose, including commercially, provided it is unaltered and attributed.

The **forms and schemas** — Annex 1's Filing Plan form, the record schemas Annex 5 names, and any conformance vectors published with this standard — are separately available under a permissive licence allowing modification, so that an implementer may lift them into code.

**A standard works only where every party is reading the same text**, and a licence permitting modified redistribution under the same name would end that. The parts meant to be built from carry no such restriction.

**0.3 · Third-party rights.** This standard has been drafted without a patent search, and **PSYCHE makes no representation that implementing it infringes no third party's rights**.

One dependency is named because a reader should know of it. The measurement primitive Annex 6 relies on for three V1 fields is third-party work by named authors [ATC], published as a preprint. **PSYCHE holds no licence to it, has sought none, and has received no statement from its authors about patents.** A Profile placing those three fields at V1 relies on that work, and the party doing so is the party exposed.

Where a reader knows of intellectual property that would be infringed by implementing this standard, saying so to PSYCHE lets the next version name it. **A standard that conceals a known encumbrance is worth less than one that discloses it.**

## 1 · Scope

SQCD-1 applies to a delivery obligation for compute capacity measured over a recurring window and paid in money.

It covers the record. It does not set availability targets, prices, remedies or credit schedules. Those are in the agreement, and this standard requires only that they be identified and fixed before the window opens.

It does not apply to the customer's own workload.

## 2 · Why a record and not an observer

Two parties dispute a month. Each holds logs, each set is internally consistent, and neither can compel the other to accept its numbers. Six weeks later a figure is agreed on a call, and nobody writes down how it was reached.

Other markets settle this without asking anyone to be disinterested. A bill of lading is issued by the carrier, who is party to the contract of carriage and has an evident stake in what it says, and banks release payment against it under letters of credit all the same — and the bank has checked nothing about the cargo. It examines the document's face and disclaims responsibility for its genuineness and accuracy [UCP600, Art. 34]. What makes the document worth paying against sits in transport law rather than banking law: **a false bill costs the carrier more than a true one.**

Statute fixes what the carrier must say and lets it decline the rest. Marks, quantity and weight are stated as the shipper furnishes them and warranted by the shipper; condition is stated as the carrier can see it; and the carrier need state nothing it has no reasonable means of checking, which is why bills read *said to contain* [HVR, Art. III rr. 3, 5]. Once said, the statement stops being the carrier's to revise. It is prima facie evidence between carrier and shipper, and **proof to the contrary becomes inadmissible once the bill reaches a third party acting in good faith** [HVR, Art. III r. 4].

Carriers do sometimes sign clean against a letter of indemnity when they know better. Where that is deliberate the courts have held the misstatement and the indemnity procured for it good for nothing, and the carrier answers to whoever relied on the bill [BJPD]. **The doctrine binds the signer whatever the signer knew**, so it works on a party with an interest.

Three properties do the work, and each of them assumes an interested carrier. Form is agreed beforehand, so the issuer cannot choose afterward what to set down and what to omit; the issuer signs, so a false statement becomes its own attributable admission rather than an argument about what anyone remembers; and the hour of signing is established by someone other than the signer, so neither of the first two can be defeated by a convenient clock.

The reservations carry as much weight as the attestation, and this standard reproduces both. A8.10 places every Grade Class field at the tier its filer can actually back: **V3 is *apparent order and condition*** — read through an interface the provider operates — and **V4 is *said to***

***contain***, carried by nothing but the signer's name under A8.11.

Current practice in this market runs the other way, and says so plainly. A practitioner guide to cloud service agreements describes the credit process as one in which the customer supplies proof of the outage and **"the CSP reviews the claims and makes a final, unilateral judgment on service credits"** [CSCC, p. 20]. The party that owes the money decides the claim. F1 and F2 exist to separate those two acts.

This standard does not require a disinterested observer of compute delivery. It requires a record with those three properties, filed by the parties themselves and checkable without either party's cooperation.

## 3 · Defined terms

**Window.** The period over which one delivery obligation is evaluated and one figure is produced.

**Measured Subject.** The unit under measurement, at the grain the contract names.

**Grade Class.** A named set of specification values under A8. Two compute hours are of the same kind when they belong to the same Grade Class.

**Layer.** One of the eight strata under B7 at which delivery can fail.

**Period Figure.** The single quantity produced for the Window by applying the declared function to the filed samples.

**Filing Plan.** The instrument under Section G by which a party declares, before its first Window, how it will produce everything this standard requires of it.

**Filing Party.** A party that files measurements. Both the provider and the customer are Filing Parties.

**Unresolved Fraction.** The share of intervals in a Window for which no Filing Party produced a conforming observation.

**Unilateral Fraction.** The share of intervals in a Window carried by exactly one Filing Party's observation.

**Divergence.** The difference between two Filing Parties' Period Figures for the same obligation and Window.

**Delivery Claim.** The object under F1. It reports.

**Counterparty Decision.** The object under F2. It decides.

## 4 · Conformance

A record is **conforming** when all of the following hold.

**4.1** Every field this standard marks MUST is present and populated.

**4.2** The Filing Plan in force at the opening of the Window is identified by digest, and the filing is consistent with it.

**4.3** The Period Figure recomputes from the filed samples under the function declared in A2, by a party holding no account with either side and using nothing but the filed record.

**4.4** The Unresolved Fraction is stated and does not exceed the ceiling declared under A6.

**The Unilateral Fraction is stated.** No ceiling applies to it.

**4.5** Every filing carries a signature under D4 and an inclusion proof under E2.

**4.6 The completeness test.** Give the record to someone who was absent and holds no account on either side. If the Period Figure, the shortfall and the amount owed all come out, and each traces to a named field, the record is complete; where a Divergence is carried to decision under Annex 2.7, each candidate figure and the amount it would owe come out, each with its basis. If any step wants a login, a phone call or a person's memory, what you have is a summary of an argument.

**4.7** A requirement marked SHOULD states what this standard expects and what a reader may assume absent a statement otherwise. **A filing that departs from a SHOULD remains conforming and MUST declare the departure under G8.** A permission marked MAY creates no obligation either way, and its exercise needs no declaration.

Completeness settles nothing. It puts one number and its basis in front of the party with authority to accept it.

# Part I — The obligation

## A · Contract Standard

*Evidence takes its meaning from the standard in force, so the standard travels with the record.*

**A1 · Obligation reference.** The clause of the agreement this record answers to.

**A2 · Metric definition.** The quantity, its unit, and the function that turns samples into a period figure. Two parties averaging differently will produce different months from identical samples.

The function MUST be one of those in Annex 2, identified by its number. A party that requires a function not in Annex 2 MUST state it in full in its Filing Plan, with a worked example over at least one hundred intervals.

**A3 · Measurement window.** Start, end, offset, and the treatment of samples straddling either edge.

**A4 · Exclusions.** Conditions that do not count against the obligation, each with the evidence required to invoke it. An exclusion invoked after the window closes is an argument.

**A5 · Tolerance and threshold.** The shortfall that contributes nothing, and the level at which the obligation is missed.

**A6 · Materiality and the unresolved ceiling.** Two numbers, both fixed before the window.

The **materiality threshold** is the divergence below which two Filing Parties' figures are treated as one.

The **unresolved ceiling** is the Unresolved Fraction above which the Period Figure is not

conforming, whatever its value.

The Profile MUST fix both: the materiality threshold by selecting a form from Annex 2.2 and supplying its parameter, the unresolved ceiling by selecting a form from Annex 2.3 and supplying its parameter. **SQCD-1 sets no default for either.** A Profile that omits either cannot be armed.

**A7 · Remedy schedule reference.** The conversion from shortfall to money, held where the evaluation cannot read it.

**A8 · Grade Class.** The kind of compute hour the obligation is for.

A Grade Class is a named tuple over the fields below. Every field carries a value or the token `not-material`. **A blank is not "unimportant"; a blank means the Grade Class is incomplete**, and an incomplete Grade Class MUST NOT be armed.

### A8.1 · Accelerator

| Field | Expression | Unit |
|---|---|---|
| `model` | manufacturer's printed designation | string, as printed |
| `form_factor` | SXM / PCIe / NVL / other | enum |
| `memory_capacity` | per accelerator | GB |
| `memory_type` | HBM3 / HBM3e / other | enum |
| `memory_bandwidth` | per accelerator | TB/s |
| `die_topology` | single- or multi-die, with partition | structure string, e.g. `2-die, 74/74 SM` |

### A8.2 · Intra-node interconnect

| Field | Expression | Unit |
|---|---|---|
| `link_generation` | NVLink generation or equivalent | enum |
| `link_bandwidth_per_gpu` | bidirectional, per accelerator | GB/s |
| `coherent_domain_size` | accelerators in one coherent domain | count |

### A8.3 · Inter-node fabric

| Field | Expression | Unit |
|---|---|---|
| `fabric_type` | IB (generation) / RoCE / other | enum |
| `fabric_bandwidth_per_gpu` | per accelerator | Gb/s |
| `allocation_contiguity` | `contiguous` / `best-effort` / `unspecified` | enum |

### A8.4 · Host

| Field | Expression | Unit |
|---|---|---|
| `host_threads_per_gpu` | processor threads ÷ accelerators | count |
| `host_memory_per_gpu` | GB ÷ accelerators | GB |
| `local_nvme_per_gpu` | capacity and sequential throughput floor | TB, GB/s |

### A8.5 · Attached storage

| Field | Expression | Unit |
|---|---|---|
| `storage_throughput_floor` | per node or per accelerator; which one MUST be stated | GB/s |
| `storage_iops_floor` | as above | IOPS |

### A8.6 · External network

| Field | Expression | Unit |
|---|---|---|
| `egress_floor` / `ingress_floor` | per node | Gb/s |

### A8.7 · Locality

| Field | Expression | Unit |
|---|---|---|
| `region` / `availability_zone` | provider's naming | string |
| `power_region` | grid balancing area | string |
| `placement_granularity` | `rack` / `hall` / `site` / `region` | enum |

### A8.8 · Tenancy

| Field | Expression | Unit |
|---|---|---|
| `tenancy` | `dedicated` / `shared` | enum |
| `preemptibility` | `reserved` / `preemptible` | enum |
| `isolation_attested` | confidential-computing mode enabled and the attestation verified | enum |

### A8.9 · Software floor

| Field | Expression | Unit |
|---|---|---|
| `driver_floor` / `firmware_floor` / `runtime_floor` | version at or above which the obligation is measured | version |

A rate quoted without a Grade Class prices an unnamed good. Two providers printing the same number are selling different things when one of them ships a downlevel driver or a fabric with half the bandwidth, and both are telling the truth.

**A8.10 · Verifiability.** Every field carries a verifiability tier. The tier is a property of the field, not of the parties, and it decides what the Profile must require in evidence.

| Tier | Definition |
|---|---|
| **V1 · Physically measurable** | The customer establishes the value by measurement, on the allocation it holds, **without the provider's cooperation and without a vendor key**. Misreporting requires defeating a physical measurement. |
| **V2 · Attestable** | A hardware root of trust signs the value and the customer verifies the signature against a manufacturer's certificate chain. Requires vendor attestation support and provider enablement. The customer verifies an endorsement, not a physical property. |
| **V3 · Observable, provider-mediated** | The customer reads the value through an interface the provider operates — driver, hypervisor, control plane, cgroup. **Detects error and misconfiguration; does not survive an adversarial provider.** |
| **V4 · Declared** | No procedure in Annex 7 reaches the value and no attestation carries it. The field is established by a signed statement under A8.11. |

Memory capacity as the driver reports it is V3: a number the provider's process tells you. **Memory bandwidth as measured is V1: a physical quantity the provider cannot fake without supplying it.**

| Field | Tier |
|---|---|
| `memory_bandwidth` · `link_bandwidth_per_gpu` · `coherent_domain_size` · `fabric_bandwidth_per_gpu` · `local_nvme` throughput · storage floors · network floors | **V1** |
| die identity · `die_topology` class · coarse geographic position | **V1**, on an unprivileged measurement primitive. See Annex 6. |
| `model` · `isolation_attested` | **V2**, where the platform's confidential-computing mode is enabled. The device's signed attestation report carries a hardware-model claim, and the report's firmware measurements are compared against manufacturer-published reference values. See Annex 6. |
| `memory_capacity` · `memory_type` · `link_generation` · `fabric_type` · `host_*` · `region` / `availability_zone` names | **V3** |
| `driver_floor` / `firmware_floor` / `runtime_floor` | **V2** where the attestation report's measurements are checked against the manufacturer's published reference values; **V3** otherwise |
| **`allocation_contiguity`** · **`tenancy`** · **`preemptibility`** · rack-level position | **V4** |

**A8.10.1 · What the tiers measure, and what they run together.** A tier answers three questions at once: who produces the evidence, whether a manufacturer vouches for it, and whether anyone independent appraises it. Those come apart. A provider that cryptographically signs its own control-plane report is producing better evidence than one that does not, and this ladder scores both as V3 because neither carries a manufacturer's endorsement. **A Profile that wants the distinction MUST state it as a required evidence class rather than expect the tier to carry it.**

The difference that survives every decomposition is the one between V1 and everything below it. **V1 asks the customer to trust a physical property. V2 asks it to trust a manufacturer. V3 asks it to trust the counterparty. V4 asks it to trust a signature and a date.**

**A8.11 · The V4 rule.** A Grade Class field at tier V4 **MUST** carry a required evidence class under the Profile.

**Where a dimension cannot be measured, this standard requires the party that knows to sign a statement of it, before the window opens, under the A10 digest.**

Latency gives a distance ordering rather than a physical position, and a customer who bought a contiguous domain and received a scattered allocation has no procedure here that establishes it.

**`tenancy` requires more care than one word can carry, and this standard separates three claims.** Whether the customer holds a die exclusively, across time, remains unestablished by any mechanism known to this standard, and stays V4. Whether the customer's work is cryptographically isolated from everything else on the machine, including the operator, **is attestable today** on parts whose confidential-computing mode is enabled, and this standard carries it as the separate field `isolation_attested` rather than folding it into `tenancy`. And whether another tenant is in fact present has been shown to be detectable from inside a tenancy, without privilege and without the provider's help, by published and peer-reviewed work on co-residency detection and on contention across a shared interconnect. **Those results are an existence result and not a deployed evidence class**: no procedure in Annex 7 implements them, and none carries the declared conditions and uncertainty bound this annex requires of a reading. A Profile MUST treat `tenancy` as V4 until one does.

The carrier is asked to sign early, on a form fixed in advance, to something it cannot later

revise. **An exclusivity declaration is the same instrument: fixed before the window, attributable, and — when it turns out to be false — a signed document that is false, rather than an argument about what anyone remembers.**

**A9 · Substitution and netting.** What may be set against what.

**A9.1** Surplus on one Measured Subject MUST NOT offset deficit on another, unless the agreement names the aggregate as the Measured Subject under B3.

**A9.2** Hours of different Grade Classes MUST NOT be netted or substituted without an adjustment factor selected from Annex 2.6 and fixed before the window.

**A9.3** Where the agreement permits aggregation, the record MUST carry both the aggregate figure and the per-subject decomposition. An aggregate that cannot be decomposed can only be accepted or refused.

This is where two contracts that both promise the same percentage stop meaning the same thing. One counts per accelerator; the other counts across a purchase order. Both instruments read correctly, and a customer can lose a rack without the number moving.

**A10 · Standard digest.** A single commitment over A1 through A9, carried in an event that precedes the Window.

Every other clause in this Section describes a term. This one describes the moment the terms stop moving. Without it, any of the nine can be edited under an unchanged version number, and the edit rewrites the standard for Windows already closed. The record would carry no trace of it, because each filing would still be consistent with the standard as it then read.

Every declaration made before the Window is signed under A10. A10 also makes the bill-of-lading argument in Section 2 available to this standard rather than merely cited by it: form agreed beforehand is form committed beforehand, and a commitment is a digest.

## Part II — The record

### B · Measurements

*Both sides file. Operator telemetry and customer-side records arrive as separate submissions and stay separate.*

**B1 · Channel.** The obligation this series measures.

**B2 · Samples at the recorded interval.** The series as the instrument produced it, ahead of any summary. A monthly figure that resists decomposition can only be accepted or refused.

**B3 · Measured subject.** The unit under measurement, at the grain the contract names: node, host, rack, cluster, endpoint. A customer who bought a rack and a clause that measures a host are describing different objects, and both instruments can be read correctly at the same time.

**B4 · Measuring party and instrument.** The party that took the reading, and what it read from.

**B5 · Clock source and stated accuracy.** Two logs that disagree about when will disagree about how much.

**B6 · Aggregation applied before filing.** Any downsampling between the instrument and this record, and the function it used.

**B7 · Layer.** The stratum at which this series observes, from the following list.

| | Layer | Observes |
|---|---|---|
| L1 | Facility | power, cooling, space |
| L2 | Host | node reachable, firmware, thermal state |
| L3 | Accelerator | present, healthy, at declared clocks |
| L4 | Interconnect | intra-node domain and inter-node fabric |
| L5 | Storage | attached volumes reachable at the declared floor |
| L6 | Network | ingress and egress at the declared floor |
| L7 | Control plane | work can be placed and started |
| L8 | Tenant workload | the customer's own code |

**B7.1** The obligation MUST name the Layers it covers.

**B7.2** An interval is delivered only where **every covered Layer was simultaneously conforming**. A conforming reading at one Layer is not an answer about another.

**B7.3** L8 MUST NOT count against the provider.

**B7.4** Where more than one covered Layer failed in an interval, the record MUST name each. Attribution then follows A4.

A machine can hold its benchmark and still deliver nothing, because the scheduler never placed the work. Measuring the accelerator and reporting the month is measuring one Layer and reporting eight.

**B8 · Minimum customer-side observation.** What a customer must observe to have standing on an interval.

**B8.1** A customer that files nothing observes nothing, and an obligation measured only by the party that owes it is measured by one party.

**B8.2** The agreement MUST name the minimum customer-side series. In the absence of a named minimum it is: reachability of each Measured Subject at an interval no coarser than the provider's, and the customer's own record of work submitted, started and completed.

**B8.3** A customer that does not meet B8.2 for a Window may still file. Its filing is recorded, and Section H applies to it as to any other.

**B8.4 · Grade verification.** Where the obligation is stated in a Grade Class, the minimum also includes a reading of each **V1** field of that class, taken under a named procedure in Annex 7·

**B8.5 · Cadence.** A V1 reading is taken at commencement, and thereafter at the cadence the Profile fixes. The commencement reading establishes that the Grade Class contracted for was the Grade Class delivered. Later readings establish that it stayed delivered.

## C · Missing Intervals

*Disagreement concentrates where one party has data and the other does not. An interval with no observation enters the record as a signed statement.*

The canonical form of a gap is the **BoundaryRecord** of the reference implementation; the fields below name its parts.

**C1 · Interval.** Start and end of the period without data. A gap that is instantaneous rather than extended is recorded as a point.

**C2 · Cause.** A code, not free text, drawn from the controlled vocabulary: `instrument-down`, `instrument-failure` (a measurement run that produced no reading, Annex 7.0.3), `retention-expired`, `collector-restarted`, `never-instrumented`, `scope-excluded`. A cause outside the vocabulary is a Profile defect.

**C3 · Attribution.** The party whose evidence is absent, and the subject the gap is against.

**C4 · Scope, materiality and signature.** The stratum the gap falls in — channel, external receipt, spot check, or global — its materiality, and a signature over C1 through C4. **An unsigned gap is a blank, and a blank is filled by whichever party argues it better.**

**C5 · Value assigned.** What number the interval carries into the Period Figure.

**C5.1** Where one Filing Party observed the interval and the other did not, the record carries the observation that exists, marked **unilateral**, and the interval counts toward the Unilateral Fraction. Whether a unilateral observation determines the interval, or produces an evidence fault under J1, follows the Profile's Annex 2.5 selection for that obligation. **SQCD-1 sets no default.**

**C5.2** Where no Filing Party observed the interval, and every Filing Party whose Filing Plan required the observation declared the gap under C1 through C4, the interval is **Unresolved**. An Unresolved interval MUST be excluded from both the numerator and the denominator of the Period Figure, and MUST be counted in the Unresolved Fraction.

**C5.3** Where the Unresolved Fraction exceeds the ceiling in A6, the Period Figure is not conforming, and no amount is owed on it until the parties agree a figure by a route outside this standard.

**C5.4** An interval that a Filing Party was required by its own Filing Plan to observe, and neither observed nor declared under C1 through C4, is an omission of that party: it is attributed to that party under C3, and J1 applies. C5.4 is read before C5.2: a declared gap is an account of what was not observed, an undeclared one is an omission. Where more than one Filing Party was so required and did not declare, each bears its own fault. What the interval then carries is assigned under C5.5.

**C5.5 · Determination.** Every interval falls in exactly one row.

| Observed by | Declared under C1 through C4 | Value assigned | Fault under J1 |
|---|---|---|---|
| both parties | — | the observations; Section H where they diverge | none |
| one party | — | C5.1: the observation that exists, marked unilateral, per the Profile's Annex 2.5 selection | the non-observing party, where its Filing Plan required the observation and it did not declare |

| Observed by | Declared under C1 through C4 | Value assigned | Fault under J1 |
|---|---|---|---|
| no party | every required party | C5.2: Unresolved; excluded from numerator and denominator, counted in the Unresolved Fraction | none |
| no party | some required party did not | the Profile's Annex 2.5 selection, read under 2.5.2 | each party that did not declare |

Counting an interval in the Unresolved Fraction states that no observation reached the record; a fault recorded beside that count says why. The count never reclassifies the fault, and the fault never supplies a reading.

## D · Provenance

*Every filing is attributable to one party, and the record names the class of evidence it came from.*

**D1 · Filing party and signing key.** One key per party. Nothing is filed on another party's behalf.

**D2 · Source class.** Operator-side, customer-side, hardware-attested, third-party-operated.

**D3 · Independence.** Whether two required sources share a trust root. Two readings from one root are one reading.

**D4 · Signature over the filed set, algorithm named.** Alteration after filing becomes detectable rather than deniable.

## E · Temporal Binding

*Everything above depends on when a record was made, and an interested party can claim any hour it likes.*

**E1 · Signing time as claimed.** It is necessary, and worth nothing by itself.

**E2 · Inclusion proof against a published log head.** A record inside a head published at time T existed before T, whatever its own clock asserted.

## F · Claim and Decision

*The two objects have different signers. One reports, the other decides.*

**F1 · Delivery claim.** Units, window, standard, and evidence set. It carries no verdict and no settlement status.

**F2 · Counterparty decision.** Accept, reject, dispute, or abstain. Signed, attributable, and bound to the object it turns on. This field alone produces `accepted`, and a reject or a dispute filed here is terminal at once; `expired` comes from the closing of the review window or the timeout at the Case deadline, as I.1.1 derives it, and from nothing else.

**F3 · Authority reference.** The grant F2 was signed under, and the limits of that grant. A signature given outside its grant binds no one, and the party that discovers this discovers it late.

# Part III — Filing, divergence, and finality

## G · Filing Plan

*Everything above is declared before the first window, by the party that will produce it.*

A Filing Party MUST file a Filing Plan before its first Window. A filing made under no Plan is not conforming.

**G1 · Filing party and role.** Provider or customer, and the obligations this Plan covers.

**G2 · Grade Classes.** Each Grade Class under A8 this Plan produces records for, with its values.

**G3 · Instruments and Layers.** For each covered Layer under B7: the instrument, what it reads from, the sample interval, the clock source, and its stated accuracy.

**G4 · Collection, validation and aggregation.** How readings travel from the instrument to the filing, including every validation, estimation or editing step and the function applied at each.

**G5 · Period function.** The Annex 2 number, or the function in full where Annex 2 does not contain it.

**G6 · Minimum observation.** For a customer, how B8 is met. For a provider, the coverage its instruments achieve across the Measured Subjects.

**G7 · Audit and test schedule.** When instruments are tested, by whom, and against what.

**G8 · Declared modifications.** Every departure from this standard, section by section. An undeclared departure makes every filing under the Plan non-conforming.

**G9 · Plan digest and effective window.** A digest over G1 through G8, and the first Window it governs.

**G10 · Change control.** A modified Plan takes effect at the opening of a Window and never inside one. A party may file under its existing Plan while a modification is pending, so long as it remains conforming under the version in force. A Plan digest that changes without a new effective Window rewrites months already closed.

## H · Divergence

*Electricity must balance, so a discrepancy surfaces on its own. Compute balances nothing. What is not filed is not noticed.*

**H1 · Duty to file regardless of agreement.** Each Filing Party MUST file for every Window, whether or not it believes there is anything to dispute. A record produced only when someone complains measures complaints.

**H2 · Divergence computed.** For each obligation and Window, the difference between the Filing Parties' Period Figures, computed by either party from the filed record.

**H3 · Concordance.** Where Divergence is at or below the materiality threshold in A6, the record states concordance and names the figure that carries forward.

**H4 · Divergence Event.** Where Divergence exceeds the threshold, a Divergence Event MUST be filed. It carries:

1. both Period Figures;
2. the interval-level decomposition of each;
3. **the first interval at which the two series part**;
4. the Layers and Measured Subjects at which they part;
5. each party's Unresolved Fraction.

**H5 · No complaint required.** A Divergence Event arises from the arithmetic. Neither party requests it, and neither party can decline it.

Localising a disagreement to its first interval ends most of them. Two parties who cannot argue about a month can usually agree about a minute.

## I · Timetable and Finality

*A record that never closes is a record that never settles. Two clocks run, because two audiences read.*

**I.1 The invoice clock.** Days are business days after the close of the Window, unless the agreement states otherwise.

| By | Event |
|---|---|
| W+2 | Each Filing Party files measurements under B and C. |
| W+5 | Divergence computed under H2. Any Divergence Event filed. |
| W+7 | Delivery Claim filed under F1. |
| W+12 | Counterparty Decision due under F2. Absent a Decision, I.1.1 applies. |
| W+30 | **Correction window closes.** Either party may file corrected measurements until this day. A corrected record yields a corrected figure, signed and filed in its own right. The Claim decided at W+12 and the Case it opened stand as decided; the difference the correction establishes is claimed by the party it favors as a credit Claim under the agreement's credit terms, or recorded as an amendment both parties sign. |
| W+45 | **Quantity finality.** No measurement may be filed or amended for the Window after this day. A measurement filed later is not part of the record. |
| W+90 | **Money finality.** The Period Figure and the amount owed are final between the parties. |

**I.1.1 Silence.** Where no Counterparty Decision is filed by the review deadline, the Case does not become accepted. Silence resolves the Case through the terminal state `expired`, derived from the event that shows the deadline passed, the closing of the review window or the timeout at the Case deadline, and never through implicit acceptance.

What expiry costs the silent party is a term of the agreement. The Profile MUST fix it by selecting from Annex 2.4. **SQCD-1 sets no default.**

A handler that reads silence as consent manufactures the one thing this standard exists to keep unmanufactured. Acceptance is an act by a party holding standing. An authority that was not exercised was not exercised.

**I.1.2** A Decision to dispute filed by the review deadline opens the correction window early. It does not move quantity finality, and it does not reopen the disputed Case: the corrected figure enters the record as the W+30 row states.

**I.1.3** After money finality the figure moves only where both parties file a signed amendment

naming this section.

**I.1.4 · Recomputation passes.** The timetable above runs one correction window. A Profile MAY state further passes between the Claim and quantity finality, each with its own date, each producing a Period Figure signed and filed in its own right.

**I.1.4.1** Where more than one pass is stated, **every Period Figure carries a pass identifier**, and a later pass supersedes an earlier one for the invoice while leaving it in the record. A figure without a pass identifier is the figure of the single pass in I.1.

**I.1.4.2** Quantity finality under I.1 binds the last pass. Passes are a way of reaching that date with better data, not a way of moving it.

Every settlement regime this standard was built from recomputes on several published dates rather than once — one grid operator runs four passes inside a hundred and five days, another runs a sequence stretching past a year, and the regime SQCD-1 takes its name from issues an initial statement and up to four recalculations. **The single-pass timetable in I.1 is a choice about compute's operating tempo, not the field's practice**, and a Profile whose obligation collateralises a facility will probably want more than one. A multi-pass regime demands in return that a reader holding a Period Figure can tell which one it is, and I.1.4.1 supplies that.

**I.2 The covenant clock.**

| By | Event |
|---|---|
| Q+10 | **Period Attestation.** A named officer of each Filing Party affirms in writing that the Filing Plan in force was implemented and followed for every Window in the quarter, and identifies every Window in which it was not. |
| Annual | **Self-audit.** Each Filing Party audits its own instruments and the path from instrument to filing, against its Filing Plan, and files the result. A party may engage an independent qualified firm to conduct it. |

**I.2.1** A Period Attestation MUST NOT be given for a quarter in which the party failed to file under H1 for more than the unresolved ceiling in A6, measured across the quarter's Windows.

The invoice clock is for the two parties. The covenant clock is for whoever lent against the contract, and the quarterly instrument exists because a lender testing coverage every quarter cannot read a monthly invoice thread.

## J · Consequences

*This Section fixes the filing timetable and the allocation of evidence faults. No fine is available here, and none is needed: what a late filing costs is the benefit of the doubt.*

SQCD-1 imposes no penalty and collects nothing. Every consequence below is evidentiary. Where parties want a monetary consequence, it belongs in A7.

**J1 · Evidence-fault allocation.** A Filing Party that, for an interval its Filing Plan required it to observe, files neither conforming measurements nor a declaration under C1 through C4 by W+2 produces an evidence fault on that interval.

Who bears an evidence fault is allocated by the Profile, **per obligation**, from Annex 2.5. **SQCD-1 sets no default and states no global rule.**

The allocation decides whether withholding pays. If a fault suspends payment, a buyer gains

by suppressing its own scheduler records; if a fault admits the claim, an operator gains by suppressing its telemetry. The incentive does not vanish by going unstated. It moves to whichever party first notices the standard was silent.

**J2 · Late filing.** A measurement filed after W+2 and before W+30 enters the record marked late. It is used. It does not displace an observation already filed on time where the two conflict, unless the counterparty accepts it.

**J3 · After quantity finality.** A measurement filed after W+45 is not part of the record and MUST NOT be used to compute the Period Figure.

**J4 · Persistent failure to file.** Where a Filing Party fails to file under H1 for more than the unresolved ceiling across a quarter's Windows, its Filing Plan is deemed not implemented for that quarter and I.2.1 applies.

**J5 · Undeclared departure.** A filing made under a Plan with an undeclared departure under G8 is not conforming, and every Period Figure resting on it is not conforming, for as long as the departure stood.

## K · Custody

**K1 · Retention.** A Filing Party MUST retain the filed record, and the samples underlying it, for thirty-six months after money finality for the Window.

**K2 · Access.** Each Filing Party holds its own filings and receives the counterparty's filed set. Nothing in this standard requires a party to disclose telemetry beyond the samples it filed.

**K3 · Onward disclosure.** A Filing Party may grant a third party access to the filed record. A grant names the party, the Windows, and its duration. A record disclosed under a grant carries its signatures and inclusion proofs, and remains checkable in the hands of the grantee without either original party's cooperation.

**K4 · Confidential terms.** Where the agreement withholds a value this standard requires to be fixed — a threshold, a remedy schedule, a Grade Class dimension — the Filing Plan MUST record that the value is fixed, identify it by digest, and name who holds it. A withheld value that is committed before the window still binds. A withheld value that is not committed is not a standard, and a Period Figure resting on one is not conforming.

K4 is the whole of what confidentiality may cost. A party may keep its numbers private. It may not keep them movable.

**K5 · Reliance by a grantee.** Where a third party has taken a grant under K3 and relied on a filing that carries a valid signature under D4 and an inclusion proof under E2, **the Filing Party that made that filing may not, as against that third party, rely on a fact contrary to it** — unless the Filing Party shows the grantee had notice of the filing's non-conformity at the time it relied.

## L · Versions

*A standard with no amendment procedure is a document. A standard whose amendments reach backwards is worse than none.*

**L1 · Identity.** This standard is identified by number and version. A Profile, a Filing Plan and a conforming record each name the version they were armed under.

**L2 · Lifecycle.** Every enumerated entry in Annex 2 carries `class` (`canonical` or `experimental`), a `rolloutState`, and a `retirementPolicy`. The standard itself carries the same three. The vocabulary is taken from the metric registry, so an entry and the standard that contains it age in the same terms.

**L3 · Effect.** A new version takes effect at the opening of a Window and never inside one.

**L4 · Non-retroactivity.** A new version does not reach a Profile already armed under an earlier one. Retirement of an enumerated entry does not reach Windows already closed under it.

**L5 · Permanent conformance. Conformance is to a named version, permanently. A later version never revokes conformance already established under an earlier one.**

**L6 · Defects found between versions.** A defect in this standard is reported to PSYCHE and enters a public errata record. Each entry names the clause, the defect, and its disposition.

**L6.1** A defect that changes what a clause requires is **substantive**, and it is corrected only by a new version under L3. **The text of a published version is never edited in place.** A10 exists so that a standard stops moving, and an errata process that rewrote a published clause would reopen the hole A10 closes.

**L6.2** A defect that changes no requirement — a typographical error, a broken cross-reference, an arithmetic slip in an example — is **editorial**, and it is corrected by a dated erratum published against the version's digest and carried alongside it. The version's own digest is unchanged, and a record armed under that version stays armed under it.

**L6.3** The errata record is published with the standard and is part of it. **A reader holding a version without its errata holds an incomplete document**, which is why L1 requires a Profile to name the version rather than to reproduce the text.

**L7 · Amendment across armed Profiles.** A Profile armed under one version continues under it, by L4. Where the parties to an armed Profile want a later change without waiting for their next arming, PSYCHE publishes the change as a digest-identified **Amendment Notice**, and the two Filing Parties adopt it by jointly signing an **Adherence** naming their existing Plan digest under G9 and the Notice's digest.

**L7.1** An Adherence takes effect at the opening of a Window and never inside one, per L3, and MUST be committed before that Window under A10.

**L7.2** An Adherence binds the two parties that signed it and reaches no other Profile. A Profile that has not adhered continues exactly as before. **Conformance already established is untouched**, because the Adherence changes what a future Window is armed under and leaves closed Windows to be read under the digest that governed them.

**L8 · Continuity.** L5 promises that a conformance question has one answer permanently, and I.2 contemplates a record read across the term of a facility. **Both promises outlive any single maintainer.** PSYCHE therefore deposits, with each version, the full text, the errata record, the schemas Annex 5 names, the registry entries Annex 2.1 indexes, and the material required to verify the digests, in a form that remains retrievable if PSYCHE does not. A version's deposit location is published with it.

L5 is not the claim that an old record satisfies the current standard. It will not; the vocabulary and the permitted sets move. It is the claim that the question *was this record conforming under SQCD-1 v1.0* has one answer, and that answer does not change when v2 issues.

# Annexes

## Annex 1 · Filing Plan form

The fields of Section G, in order, as a form. A Plan filed on this form and signed under D1 satisfies G.

```
SQCD-1/P · FILING PLAN

G1  Filing party ................ [ legal name ]
    Role ........................ [ provider | customer ]
    Obligations covered ......... [ A1 references ]
    First Window governed ....... [ date ]

G2  Grade Classes ............... [ per A8.1–A8.9, one block per class ]

G3  Instruments and Layers ..... [ per covered Layer: instrument,
                                   source, interval, clock, accuracy ]

G4  Collection and validation .. [ each step, each function ]

G5  Period function ............ [ Annex 2 number, or full statement ]

G6  Minimum observation ........ [ B8 for a customer; coverage for a
                                   provider ]

G7  Audit and test schedule .... [ cadence, party, standard ]

G8  Declared modifications ..... [ section, departure, reason ]
                                 [ "none" is a permitted entry ]

G9  Plan digest ................ [ algorithm, digest ]

Signed ......................... [ D1 key identifier ]
```

## Annex 2 · Enumerated sets

Every set below enumerates **forms**, not values. The Profile selects a form by number and supplies its parameter. Comparability across contracts comes from knowing which form was selected; it does not require that everyone select the same one.

Where a term allocates loss between the parties, this standard enumerates what may be chosen and assigns no value.

**2.1 · Period functions.** A2 requires the function to be **named**, not described. A name resolves to an entry in the metric registry.

**2.1.1 · Relationship to ISO/IEC 19086-2.** A registry entry is a cloud service metric under **ISO/IEC 19086-2:2018, Clause 8**, extended under that clause's own extension provision at 8.1.4. It carries the classes and attributes Clause 8 defines — `Metric`, with its identifier, description, source, scale, expression, parameters, rules and underlying metrics, together with the `Expression`, `Parameter` and `Rule` classes those refer to — and a conforming entry may be serialised against the normative XML schema that document supplies.

`scale` is taken from 19086-2, and its binding there is preserved here: a cloud service level objective takes an interval or ratio scale, and a cloud service qualitative objective a nominal or ordinal one. **Every function in this annex is quantitative and takes an interval or ratio scale.** The Grade Class fields under A8 are nominal and ordinal, so they are qualitative objectives rather than level objectives; A8.10 governs them, and no function in this annex applies to them.

**2.1.2 · What this standard adds.** Clause 8 specifies a metric well enough to describe a service and to compare two of them. Settlement asks three further questions, and a registry entry under this standard answers them.

| Added to the 19086-2 entry | Why settlement requires it |
|---|---|
| **`missingDataBehavior`** | What the function does with an interval no sample reached. This value is exercised in Section C. A function that leaves it unstated has no answer for the case Section C exists for. |
| **`determinismConstraints`** | Whether two parties running the entry over one filed sample set are obliged to reach one figure. Comparability is enough to describe a service; settlement needs identity. |
| **`version`** · **`class`** · **`rolloutState`** · **`retirementPolicy`** · **`reviewRefs`** · **`profileScope`** | ISO/IEC 19086-2 carries no version field. A2 is committed under the A10 digest and read back under L1 through L5, and a metric able to change under a fixed name defeats every one of those clauses. |

**A metric a Window's figure was computed under must be recoverable years later, exactly as it then read.**

**2.1.3 · Where an entry lives.** An entry is an object under the `metric_registry_entry` schema Annex 5 names. A Profile that names an entry MUST carry that entry, at the version named, among the artifacts the A10 digest is computed over, so the record of a Window holds the entry as it read when the Window was armed and an evaluation resolves the name from that commitment and from nothing outside it. The index below is not such a carrier; the seven `canonical` entries it names are deposited with this version under L8, as `sqcd1-registry/`.

**The table below is an index of the `canonical` entries, not a substitute for them.** A Profile selects by identifier; an evaluation runs against the entry, not this table.

| No. | Function |
|---|---|
| **F-01** | Time-weighted availability. Conforming interval-seconds divided by total interval-seconds in the Window, per Measured Subject, arithmetic mean across subjects. |
| **F-02** | Subject-weighted availability. As F-01, weighted by the accelerator count of each Measured Subject. |
| **F-03** | Worst-subject availability. The minimum of the per-subject F-01 values across the Window. |
| **F-04** | Aggregate availability. Conforming subject-seconds summed across all subjects, divided by contracted subject-seconds. Permitted only where the agreement names the aggregate as the Measured Subject under B3. |
| **F-05** | Delivered hours. The count of conforming Measured-Subject-hours in the Window, undivided. |
| **F-06** | Shortfall hours. Contracted Measured-Subject-hours less F-05. |

| No. | Function |
|---|---|
| **F-07** | Windowed availability with error budget. As F-01, with the first *n* non-conforming interval-seconds per subject per Window excluded, *n* declared in A5. |

F-01 and F-04 over the same samples produce different months, and that is why A2 requires one of them to be named before the Window opens.

An entry marked `experimental` MAY be selected by a Profile and MUST be recorded as experimental in the Filing Plan. Retirement of an entry does not reach Windows already closed under it; see Section L.

**2.2 · Materiality threshold forms.** The divergence below which two Filing Parties' Period Figures are treated as one. A6 requires a form and a parameter.

This set enumerates the forms the **boundary tolerance** parameter may take. It introduces no new field: the parameter already exists in the armed parameter pack, and these are the shapes it is permitted to have.

| No. | Form |
|---|---|
| **MT-01** | Fraction of the Period Figure. Parameter: the fraction. |
| **MT-02** | Absolute quantity on one Measured Subject. Parameter: the quantity and its unit. |
| **MT-03** | The greater of an MT-01 fraction and an MT-02 quantity. Parameters: both. |
| **MT-04** | The lesser of an MT-01 fraction and an MT-02 quantity. Parameters: both. |
| **MT-05** | Zero. Every divergence is material and produces a Divergence Event. |

MT-03 favours the party filing the larger figure on small Windows; MT-04 favours the other.

**2.3 · Unresolved ceiling forms.** The Unresolved Fraction above which the Period Figure is not conforming. A6 requires a form and a parameter.

As with 2.2, this set enumerates the permitted forms of a threshold parameter of the armed parameter pack, `unresolved_ceiling`, which carries the selected form and its parameter.

| No. | Form |
|---|---|
| **UC-01** | Fraction of intervals in the Window, in aggregate. Parameter: the fraction. |
| **UC-02** | Absolute count of Unresolved intervals. Parameter: the count. |
| **UC-03** | Fraction of intervals, evaluated **per Measured Subject**, the ceiling breached if any subject exceeds it. Parameter: the fraction. |
| **UC-04** | No ceiling. The Period Figure conforms at any Unresolved Fraction, **provided the fraction is stated**. |

UC-01 and UC-03 diverge exactly where the aggregate hides a single blind subject. UC-04 is permitted because some parties will want it; it does not permit leaving the fraction unstated.

**2.4 · Expiry pricing.** What silence costs the silent party. I.1.1 requires a selection.

**None of these produces `accepted`.** The Case's terminal state under every entry below is `expired`. What moves is money under a term fixed before the Window, not consent. No entry in this set may be read as acceptance, and a Profile that describes one as acceptance is not conforming.

| No. | Form |
|---|---|
| **EX-01** | No amount. Expiry closes the Case and nothing is owed on the claim either way. |
| **EX-02** | Invoice stands, remedy lapses. The underlying invoice remains payable on its own terms; the claim for remedy in this Case lapses unexercised. |
| **EX-03** | Remedy stands, uncontested. The amount computed under the remedy schedule is owed; the silent party spent its opportunity to contest. |
| **EX-04** | Stated cost. Expiry carries an amount payable by the silent party, fixed in advance and independent of the merits. Parameter: the amount. |

EX-02 and EX-03 are symmetric and both are ordinary negotiated positions.

**2.5 · Evidence-fault allocation.** Who bears an interval on which a Filing Party did not file. J1 requires a selection **per obligation**.

| No. | Form |
|---|---|
| **EF-01** | Suspended. The interval carries no value and no amount is owed on it until evidence is filed or the correction window closes. |
| **EF-02** | Admitted. The interval counts as delivered. |
| **EF-03** | Denied. The interval counts as not delivered. |
| **EF-04** | Attributed. The interval is borne by the party C5.4 attributes it to; where it carries no observation, 2.5.2 applies. |
| **EF-05** | Excluded. The interval leaves both numerator and denominator and counts toward the Unresolved Fraction. |
| **EF-06** | Split. Borne in a declared ratio. Parameter: the ratio. |
| **EF-07** | Estimated. The interval carries a value computed from a formula named before the Window, over observations the Filing Party did produce. Parameters: the formula, and whether the estimate is replaceable by actual data or final on filing. |

**EF-07 answers a different question from EF-01 through EF-06**, and a Profile may need both. The first six allocate *who bears* an interval; EF-07 states *what number it carries*. Settlement-quality metering regimes this standard was drawn from do the second thing, at least one of them by a formula published in advance rather than by a judgment made afterward, and at least one marks an estimate that will never be replaced as final on the day it is filed. A Profile selecting EF-07 without stating which of the two it means has fixed nothing.

**2.5.1 Conformance rule.** A selection is not conforming where the party that holds the required evidence is the party that gains from its absence. EF-02 with an operator-held evidence class, and EF-03 with a customer-held one, are the two standing instances. This is detectable when the Profile is armed and MUST be reported then.

2.5.1 refuses a selection on its merits only where a party is paid to stay blind.

**2.5.2 An interval no party observed.** Where C5.4 attributes an interval that carries no observation, the selected form assigns its value as follows. EF-01 suspends it. EF-02 counts it as delivered and EF-03 as not delivered. EF-04 and EF-05 exclude it from numerator and denominator and count it in the Unresolved Fraction, the fault recorded against each party C5.4 names. EF-06 applies its ratio to whatever consequence the agreement attaches to the fault under A7, the interval itself excluded as under EF-05. EF-07 computes its value from the named formula.

**2.6 · Substitution and adjustment forms.** A9.2 forbids netting or substituting hours of different Grade Classes without an adjustment factor. These are the forms that factor may take.

| No. | Form |
|---|---|
| **SB-01** | No substitution. Hours of different Grade Classes are separate obligations and never offset. |
| **SB-02** | Named-pair factor. A scalar for one ordered pair of Grade Classes, in one direction only. Parameters: the pair, the direction, the scalar. |
| **SB-03** | Field-derived factor. The ratio of one named A8 field between the two classes, that field stated. Permitted only where the obligation is measured on that field. |
| **SB-04** | Schedule. A table of factors over an enumerated set of class pairs, fixed before the Window. |
| **SB-05** | Upward only. Substitution permitted where every A8 field of the delivered class meets or exceeds the contracted class, at factor 1. |

**2.6.1** A factor MUST NOT be derived from price. A price ratio prices two goods; it does not establish that one discharges an obligation stated in the other.

**2.6.2** SB-03 and SB-05 are checkable from the Grade Classes alone. SB-02 and SB-04 are negotiated numbers and MUST be fixed under the A10 digest before the Window.

**2.6.3 · Comparison under SB-05.** A field carrying a quantity, a count or a floor meets or exceeds where the delivered value is not below the contracted value; a version under A8.9 meets where it is not older. A field carrying a name or an enumerated value — `model`, `form_factor`, `memory_type`, `die_topology`, `link_generation`, `fabric_type`, `allocation_contiguity`, `region`, `availability_zone`, `power_region`, `placement_granularity`, `tenancy`, `preemptibility`, `isolation_attested` — meets only where the delivered value equals the contracted value, unless the agreement orders that field's values under the A10 digest before the Window. A field marked `not-material` in the contracted class is not compared.

SB-05 is the only form under which a provider may substitute with no factor to compute or negotiate, and a customer verifies it by reading two rows of the registry. It is also the only form that cannot be used to deliver less.

**2.7 · Divergence resolution.** How a material Divergence under Section H is carried to one quantity. H4 requires a selection.

**None of these produces `accepted`.** Every entry below resolves a **measurement**. `accepted` continues to come from F2 alone and `expired` from I.1.1, and a Profile that describes any entry here as acceptance is not conforming. Resolving a divergence establishes measurement finality; counterparty finality is a separate act by a party holding standing.

| No. | Form |
|---|---|
| **DR-01** | No resolution. Both Period Figures stand in the record and the quantity is agreed by a route outside this standard. |
| **DR-02** | Tolerance mean. Where the two figures differ by no more than a stated bound, their arithmetic mean governs. Parameter: the bound. |
| **DR-03** | Third reading, closest two. A third reading is taken by a party named before the Window; the mean of the two closest of the three governs, and the outlier is discarded without adjudication. Parameters: who takes the third reading, and who bears its cost. |
| **DR-04** | Referee. A single referee named before the Window determines the quantity from the filed record. Parameters: the referee, and the cost rule. |
| **DR-05** | Panel. An odd-numbered panel drawn from a roster named before the Window decides by majority. Parameters: the roster, the panel size, and whether one appeal lies. |
| **DR-06** | Carried to decision. The divergence is carried into F2 unresolved, and the party with standing decides on a record that states both figures and the first interval at which they part. |

**2.7.1** A selection MUST be made. **DR-01 is a selection**, and a Profile that names it has chosen to leave the quantity to a conversation. Two parties may take that position deliberately, and this standard requires them to take it on the page rather than by omission.

**2.7.2** DR-02 through DR-05 determine a quantity and MUST NOT be described as determining what is owed. The remedy schedule under A7 converts a quantity to money, and F2 settles.

**2.7.3** Where the selected form requires a third party, that party MUST be named before the Window opens, together with the basis on which it is paid. A referee chosen after the divergence is known is chosen by whoever is ahead.

Physical commodity markets answer this question and have for a century: a mechanical two-of-three average, a single referee whose cost falls on whoever turns out to be wrong, a randomly drawn panel voting by majority, an umpire appointed by the trade body when the two arbitrators cannot agree. Each of them terminates in a number. **What H4 contributes to that inheritance is the localisation** — the parties arrive at whichever of these forms they chose already knowing the interval at which their instruments parted, which none of those procedures supplies.

**2.8 · Conformity decision near a limit.** Whether a reading that sits close to a Grade Class floor or an A5 threshold counts as conforming. Annex 7 requires a selection wherever a measured value is compared to a limit.

A reading carries an expanded uncertainty under 7.0.2. Comparing it to a limit as though it were exact makes the outcome near that limit close to a coin flip, and which way the coin falls is worth money to one of the parties. **Which form applies allocates that risk, and this standard sets no default.**

| No. | Form |
|---|---|
| **CD-01** | Simple acceptance. The reading is compared to the limit directly and the risk arising from measurement uncertainty falls where it falls. |
| **CD-02** | Guarded acceptance. The reading must clear the limit by at least a stated guard band before the field counts as conforming. Parameter: the band, stated as a multiple of the expanded uncertainty $U$. |
| **CD-03** | Guarded rejection. The reading must miss the limit by at least a stated guard band before the field counts as non-conforming. Parameter: the band. |
| **CD-04** | Stated maximum risk. The band is derived so that the probability of a wrong conformity decision does not exceed a stated value. Parameter: the maximum risk. |

**2.8.1** A form MUST be selected for every Grade Class field measured under Annex 7 against a floor, and for the A5 threshold where the Period Figure is built from measured values rather than from counted intervals.

**2.8.2** CD-02 protects the party that loses by a wrong acceptance; CD-03 protects the party that loses by a wrong rejection. **They are the same instrument pointed in opposite directions**, and a Profile selecting one has decided which error it would rather make.

**2.8.3** Where a selection names a guard band as a multiple of $U$, the coverage factor $k$ used to compute $U$ MUST be the one the procedure states under 7.0.2. A band expressed against an unstated $k$ is a band of unknown width.

## Annex 3 · Worked example

One Window, one obligation, two Filing Parties. Every number below is derived from the sample counts in 3.2 and can be recomputed from them with no other input.

**What this example does not exercise.** It demonstrates counted intervals and a period function, and nothing else. **No reading under Annex 7 appears in it.** The percentages below are

counts of conforming intervals divided by counts of eligible ones — exact ratios of integers — and they carry no expanded uncertainty because no instrument produced them. A Window whose Period Figure rests on measured values instead of counted intervals runs 7.0.2 for every reading and 7.0.2.1 for every comparison against a floor, and its figure would be reported with the uncertainty those clauses require. **The six decimal places below are the arithmetic of a division, not a claim about instrument precision.** A standard that publishes rules and withholds arithmetic leaves every implementer to guess at the same question.

### 3.1 · The standard in force.

| Ref | Term | Value |
|---|---|---|
| A1 | Obligation | Availability of contracted capacity, clause 4.2 |
| A2 | Period function | **F-01**, time-weighted availability, per subject, arithmetic mean across subjects |
| A3 | Window | 2026-06-01T00:00:00Z → 2026-07-01T00:00:00Z. 720 hours. Sample interval 300 s. Samples straddling either edge are assigned to the Window containing their start. |
| A4 | Exclusions | Announced maintenance, notified ≥ 72 h in advance. Excluded intervals are removed from **both** numerator and denominator. |
| A5 | Tolerance / threshold | No error budget. Obligation missed below 99.90 %. |
| A6 | Materiality | **MT-02**, absolute quantity on one Measured Subject. Parameter: **12 intervals** (one hour). |
| A6 | Unresolved ceiling | **UC-01**, fraction of eligible intervals, in aggregate. Parameter: **1 %**. |
| A7 | Remedy schedule | ≥ 99.90 % → none · 99.85–99.90 % → 5 % · 99.50–99.85 % → 10 % · < 99.50 % → 25 %, of the Window fee |
| A8 | Grade Class | `GC-H200-SXM-141`, 8 accelerators per Measured Subject |
| A10 | Standard digest | committed 2026-05-28, before the Window opened |
| A8.11 | V4 declarations | `tenancy = dedicated`, `allocation_contiguity = contiguous`, both signed by the operator 2026-05-28, under the A10 digest |
| A9 | Substitution | **SB-01**, no substitution |
| B3 | Measured Subject | Node. Eight subjects, N-01 … N-08 |
| B7 | Covered Layers | L2, L3, L4, L6, L7. L1 and L5 out of scope. **L8 never counts.** |
| B8 | Minimum customer observation | Reachability of each subject at 300 s, plus the customer's own record of work submitted, started and completed |
| Annex 2.5 | Evidence-fault allocation | **EF-04**, attributed to the party whose Filing Plan required the observation |
| H4 | Divergence resolution | **DR-06**, carried to decision |
| I | Timetable | W+2 measurements · W+5 divergence · W+7 Claim · W+12 Decision · W+30 correction closes · W+45 quantity final · W+90 money final |
| I.1.1 | Expiry pricing | **EX-02**, invoice stands, remedy lapses |
| — | Price | $2.90 per accelerator-hour |

Window fee: 8 subjects × 8 accelerators × 720 h = **46 080 accelerator-hours** × $2.90 = **$133 632.00**

**3.2 · What the instruments recorded.** Intervals per subject: 720 h × 3600 ÷ 300 = **8 640**. Total subject-intervals: 8 × 8 640 = **69 120**.

| # | Event | Extent |
|---|---|---|
| **E1** | Staged firmware maintenance, notified 2026-05-25. **A4 exclusion.** | 24 h = 288 intervals, on N-01 … N-04 → **1 152** subject-intervals |
| **E2** | Both collectors lost their upstream, and each party filed a BoundaryRecord for the interval with cause `instrument-down`. **No Filing Party observed; both declared.** | 45 min = 9 intervals, all subjects → **72** subject-intervals |
| **E3** | Customer probe down. **Operator filed; customer did not.** | 4 h = 48 intervals, all subjects → **384** subject-intervals |

| # | Event | Extent |
|---|---|---|
| **E4** | N-03: control plane could not place work. **L3 healthy throughout; L7 non-conforming.** Both parties filed, and agree. | 6 h = 72 intervals on N-03 |
| **E5** | N-05: customer probe records unreachable at L6; operator's collector records conforming. **The parties disagree.** | 3 h = 36 intervals on N-05, from 2026-06-21T02:00:00Z |

**E4 is the case B7.2 exists for.** Every accelerator on N-03 was healthy and at its declared clocks for all 72 intervals. The obligation covers L7. An interval is delivered only where every covered Layer was simultaneously conforming, so the 72 intervals are non-conforming and a reading at L3 is not an answer about them.

**3.3 · Denominators.** Excluded (E1) is removed from both sides. Unresolved (E2), declared by both parties, is removed from both sides under C5.2; had either party left its gap undeclared, C5.4 would have attributed the interval to it instead.

| | Intervals |
|---|---|
| Total subject-intervals | 69 120 |
| less A4 exclusions (E1) | − 1 152 |
| **Eligible before C5** | **67 968** |
| less Unresolved (E2) | − 72 |
| **Eligible** | **67 896** |

Per subject: N-01 … N-04 → 8 640 − 288 − 9 = **8 343**. N-05 … N-08 → 8 640 − 9 = **8 631**. Check: 4 × 8 343 + 4 × 8 631 = 33 372 + 34 524 = 67 896. ✓

**Unresolved Fraction** = 72 ÷ 67 968 = **0.105932 %**. Ceiling 1 % (UC-01). Conforming.
**Unilateral Fraction** = 384 ÷ 67 968 = **0.564972 %**. No ceiling applies; it is disclosed.

E3 is unilateral: only the operator observed. The customer's Filing Plan required that observation under B8 and the customer did not produce it, so C5.4 applies and J1 routes to the Profile's Annex 2.5 selection. Under **EF-04** the interval is borne by the customer, and the operator's record — conforming — determines all 384.

**3.4 · The two Period Figures.** Per-subject availability, conforming ÷ eligible:

| Subject | Operator | Customer |
|---|---|---|
| N-01, N-02, N-04 | 1.000000 | 1.000000 |
| **N-03** | 1 − 72/8 343 = **0.99137001** | **0.99137001** |
| **N-05** | 1.000000 | 1 − 36/8 631 = **0.99582899** |
| N-06, N-07, N-08 | 1.000000 | 1.000000 |

**F-01, operator** = (0.99137001 + 7 × 1) ÷ 8 = **99.892125 %**
**F-01, customer** = (0.99137001 + 0.99582899 + 6 × 1) ÷ 8 = **99.839987 %**

**Divergence** = 0.052138 percentage points. All of it arises from 36 intervals on N-05.

**3.5 · Divergence.** Materiality is MT-02 at 12 intervals on one Measured Subject. The divergence is 36 intervals on N-05. **36 > 12 → material.** A Divergence Event is filed under H4, by arithmetic, without either party requesting it.

| H4 field | Value |
|---|---|
| Both Period Figures | 99.892125 % · 99.839987 % |
| Interval decomposition | filed, per subject, per interval |
| **First interval at which the series part** | **2026-06-21T02:00:00Z** |
| Layer and subject | **L6**, N-05 |
| Unresolved Fraction, each party | 0.105932 % · 0.105932 % |

The evidence set is now known to be inconsistent. Under DR-06 the evaluation does not choose between the two readings; it records the conflict and carries it forward, and returns **`requires-review`** rather than a support assertion. Choosing a winner here would be the evaluation exercising an authority it does not hold.

**Locating the disagreement to 2026-06-21T02:00:00Z is most of the work.** The parties are no longer arguing about a month. They are arguing about three hours on one node, with both instruments' readings of those three hours in front of them.

### 3.6 · The money.

| | Period Figure | Tier | Credit |
|---|---|---|---|
| Operator's filing | 99.892125 % | 99.85 – 99.90 % | 5 % → **$6 681.60** |
| Customer's filing | 99.839987 % | 99.50 – 99.85 % | 10 % → **$13 363.20** |

**The disputed amount is $6 681.60**, and it is 36 five-minute intervals on one node out of eight, in a 720-hour month.

Both quotes are computed and both are carried. **Neither is owed yet.** The remedy quote names an amount and settles nothing; the amount owed is fixed when the party with standing signs a decision under F2. Measurement finality has been reached and counterparty finality has not.

**3.7 · The same samples under a different declared function.** A2 required the function to be named before the Window opened. The table below shows why.

| Function | Operator's figure | Tier | Credit |
|---|---|---|---|
| **F-01** time-weighted, mean across subjects | 99.892125 % | 99.85 – 99.90 % | 5 % → **$6 681.60** |
| **F-04** aggregate, subject-seconds pooled | 99.893955 % | 99.85 – 99.90 % | 5 % → $6 681.60 |
| **F-03** worst subject | **99.137001 %** | **< 99.50 %** | **25 % → $33 408.00** |

Identical samples and identical instruments produce **five times the money**.

F-01 and F-04 differ here in the fourth decimal, because the eligible denominators differ only by E1's staged maintenance; on a fleet where subjects are excluded unequally the gap widens. F-03 differs by three quarters of a percentage point, because it reports the worst subject rather than the fleet, and one node was down for six hours.

None of the three is wrong. **The one that is wrong is the contract that did not say which.**

**3.8 · Completeness.** Give this record to someone who was absent and holds no account on either side.

| | Comes out |
|---|---|
| The period figure | 99.892125 % and 99.839987 %, each recomputable from 3.2 and 3.3 |
| The shortfall | 72 non-conforming intervals agreed, 36 disputed, located to the first interval at which the series part |
| The amount owed | $6 681.60 or $13 363.20, quoted under A7, **owed when F2 is signed** |
| What was not observed | Unresolved 0.105932 %, Unilateral 0.564972 %, both disclosed |
| What was not verifiable | `tenancy` and `allocation_contiguity` are V4. Neither was measured. Both were declared by the operator on 2026-05-28 under the A10 digest, and the declarations are in the record. |

Each traces to a named field.

## Annex 4 · Grade Class registry

An index of manufacturer-published specifications, each entry naming its source and the date it was read. Values are transcribed as printed. A field the manufacturer states only as a system total is marked **ABSENT** rather than divided, and a field for which the manufacturer prints two irreconcilable figures is marked **ABSENT** with both recorded.

**Listing is citation rather than certification**, and no manufacturer has undertaken anything to PSYCHE by appearing here. The V1 procedures in Annex 7 are where a customer establishes that the part in front of it matches.

This annex lists Grade Classes by name, with their A8 values, and PSYCHE maintains it.

**Only verified values are printed.** A field the manufacturer's page does not state is marked ABSENT and left empty. Arithmetic performed on printed values is not a printed value and is not entered here.

Device fields (A8.1, A8.2 where the device fixes it) come from the manufacturer. Deployment fields — `coherent_domain_size` where the rack fixes it, `fabric_*`, `allocation_contiguity`, `tenancy`, storage and network floors — are properties of a deployment, not of a device, and are supplied per Profile.

**GC-H200-SXM-141.** Source: nvidia.com/en-us/data-center/h200/ · retrieved 2026-09-02 · quotation

| Field | Value |
|---|---|
| `model` | NVIDIA H200 |
| `form_factor` | SXM |
| `memory_capacity` | 141 GB |
| `memory_type` | HBM3e |
| `memory_bandwidth` | 4.8 TB/s |
| `link_bandwidth_per_gpu` | 900 GB/s |
| PCIe | Gen5, 128 GB/s |
| TDP | up to 700 W (configurable) |

**GC-H200-NVL-141.** Same source.

| Field | Value |
|---|---|
| `model` | NVIDIA H200 NVL |
| `form_factor` | PCIe dual-slot, air-cooled |

| Field | Value |
|---|---|
| `memory_capacity` | 141 GB |
| `memory_type` | HBM3e |
| `memory_bandwidth` | 4.8 TB/s |
| `link_bandwidth_per_gpu` | 900 GB/s per GPU |
| PCIe | Gen5, 128 GB/s |
| TDP | up to 600 W (configurable) |

**GC-GB300-NVL72.** Source: nvidia.com/en-us/data-center/gb300-nvl72 · retrieved 2026-09-02 · quotation

| Field | Value |
|---|---|
| configuration | 72 NVIDIA Blackwell Ultra GPUs, 36 NVIDIA Grace CPUs |
| `coherent_domain_size` | 72 |
| GPU memory, total | 20 TB HBM3E |
| fast memory, total | 37 TB |
| NVLink bandwidth, total | 130 TB/s |
| `fabric_bandwidth_per_gpu` | 800 Gb/s (NVIDIA ConnectX-8 SuperNIC) |
| `memory_capacity` per accelerator | **ABSENT** — the page states the total only |
| `link_bandwidth_per_gpu` · rack power | **ABSENT** |

**GC-H100-SXM-80.** Source: nvidia.com/en-us/data-center/h100/ · retrieved 2026-09-02 · quotation

| Field | Value |
|---|---|
| `model` | NVIDIA H100 |
| `form_factor` | SXM |
| `memory_capacity` | 80 GB |
| `memory_type` | **ABSENT** — the specification table states capacity and bandwidth, and the type appears only in surrounding prose |
| `memory_bandwidth` | 3.35 TB/s |
| `link_bandwidth_per_gpu` | 900 GB/s |
| PCIe | Gen5, 128 GB/s |
| TDP | up to 700 W (configurable) |

**GC-H100-NVL-94.** Same source.

| Field | Value |
|---|---|
| `model` | NVIDIA H100 NVL |
| `form_factor` | PCIe dual-slot, air-cooled |
| `memory_capacity` | 94 GB |
| `memory_type` | **ABSENT** |
| `memory_bandwidth` | 3.9 TB/s |
| `link_bandwidth_per_gpu` | 600 GB/s |
| PCIe | Gen5, 128 GB/s |
| TDP | 350–400 W (configurable) |

The manufacturer's current page prints no 80 GB PCIe H100. **`GC-H100-PCIe-80` is therefore not registered**, and a contract naming one is naming a part this registry cannot resolve.

**GC-B200-SXM.** Sources: nvidia.com/en-us/data-center/hgx/ and nvidia.com/en-us/data-center/dgx-b200/ · retrieved 2026-09-02 · quotation

| Field | Value |
|---|---|
| `model` | NVIDIA Blackwell, B200 |
| `form_factor` | SXM |
| `link_bandwidth_per_gpu` | 1.8 TB/s, GPU-to-GPU |
| `memory_type` | HBM3e (DGX B200) |
| `memory_capacity` | **ABSENT** — eight-GPU totals only: 1.4 TB (HGX B200), 1,440 GB (DGX B200) |
| `memory_bandwidth` | **ABSENT** — 64 TB/s (DGX B200) is an eight-GPU aggregate |
| `coherent_domain_size` | **ABSENT** — the board is printed as 8× Blackwell SXM with 2× NVSwitch and 14.4 TB/s aggregate NVLink |
| networking | up to 400 Gb/s per port (DGX B200) · 0.8 TB/s system networking bandwidth (HGX B200) |

**Read the B200 entry against the two H200 entries.** For the older part the manufacturer prints per-accelerator memory capacity and per-accelerator bandwidth. For the current part it prints eight-GPU totals, and the two fields a delivery obligation is actually written in have no printed per-accelerator value at all.

Dividing the total by eight would supply them, and this registry declines to, because a quotient is not a quotation and a Grade Class that mixes the two stops being checkable against anything. The consequence is operational rather than editorial: for current-generation hardware, `memory_capacity` and `memory_bandwidth` reach the record either as a Profile declaration under A8, or as a reading under **MP-01**. The registry cannot supply them, and the standard's answer to a field the datasheet leaves out is the same as its answer everywhere else — measure it, or have the party that knows sign it.

**GC-MI300X · GC-MI325X · GC-MI350X · GC-MI355X.** Source: the single-accelerator product page for each part at amd.com, each cross-checked against its product brief or datasheet PDF · retrieved 2026-09-02 · quotation

| | MI300X | MI325X | MI350X | MI355X |
|---|---|---|---|---|
| `form_factor` | OAM | OAM | OAM Module | OAM |
| `memory_capacity` | 192 GB | 256 GB | 288 GB | 288 GB |
| `memory_type` | HBM3 | HBM3E | HBM3E | HBM3E |
| `memory_bandwidth` | 5.3 TB/s | 6 TB/s | 8 TB/s | 8 TB/s |
| Infinity Fabric links | 8, at 128 GB/s per link | 7, at 128 GB/s per link | 7, at 153 GB/s per link | 7 |
| host | PCIe 5.0 x16, 128 GB/s | PCIe Gen 5, 128 GB/s | PCIe 5.0 x16, 128 GB/s | PCIe 5.0 x16 |
| TDP | up to 750 W | **ABSENT** | 1000 W typical board power | **ABSENT** |
| `coherent_domain_size` | **ABSENT** | **ABSENT** | **ABSENT** | **ABSENT** |
| `fabric_bandwidth_per_gpu` | **ABSENT** | **ABSENT** | **ABSENT** | **ABSENT** |

**`link_bandwidth_per_gpu` is unresolved for two of these four.** Each platform page prints a figure labelled *Total Aggregate Bi-directional I/O Bandwidth* (*Peer-to-Peer*) — 896 GB/s for MI300X and MI325X, 1,194.8 GB/s for MI350X and MI355X — and the label says neither per accelerator nor per platform. For the first pair the manufacturer's own datasheet resolves it: it states seven links reaching the other seven accelerators at 128 GB/s between each pair, and $7 \times 128 = 896$ exactly, so the printed figure is a per-accelerator aggregate. **For the second pair nothing in the cited documents resolves it.** The same brochure prints 153.6 GB/s per

link in its table and 160 GB/s per link in its prose, and neither figure times seven reaches 1,194.8. The field stays **ABSENT** for MI350X and MI355X.

**GC-TPU-V5E · GC-TPU-V5P · GC-TPU-V6E · GC-TPU-V7.** Source: docs.cloud.google.com/tpu/docs · retrieved 2026-09-02 · quotation. **The manufacturer labels per-chip and per-pod figures itself**, and the per-chip column below is its own.

| | v5e | v5p | v6e | v7 |
|---|---|---|---|---|
| HBM per chip | 16 GB | 95 GiB | 32 GB | 192 GiB |
| HBM bandwidth per chip | 800 GiBps | 2,765 GBps | 1,638 GBps | 7,380 GBps |
| Interconnect per chip, bidirectional | 400 GBps | 1,200 GBps | 800 GBps | 1,200 GBps |
| chips per pod, maximum | 256 | 8,960 | 256 | 9,216 |
| topology | 2D torus | 3D torus | 2D torus | **ABSENT** |
| power | **ABSENT** | **ABSENT** | **ABSENT** | **ABSENT** |
| `form_factor` · `coherent_domain_size` | **ABSENT** | **ABSENT** | **ABSENT** | **ABSENT** |

**The unit this manufacturer's documentation specifies is a chip inside a pod**, and "chips per pod" is a scheduling maximum rather than a set of accelerators mutually reachable at link bandwidth. **`coherent_domain_size` has no printed value for any generation**, and a Profile contracting for these parts fixes it by declaration under A8 rather than by reference to this registry.

**GC-GAUDI3.** Source: the manufacturer's HL-325L OAM product brief · retrieved 2026-09-02 · quotation

| Field | Value |
|---|---|
| `model` | Intel Gaudi 3, HL-325L |
| `form_factor` | OCP Accelerator Module (OAM) v2.0 |
| `memory_capacity` | 128 GB |
| `memory_type` | HBM2e |
| `memory_bandwidth` | "total throughput of 3.7 TB/s" |
| networking | 24 × 200 GbE RoCE v2 RDMA ports; 9.6 Tb/s bidirectional |
| host | PCIe Gen 5.0 x16 |
| TDP | up to 900 W |
| `link_bandwidth_per_gpu` · `fabric_bandwidth_per_gpu` | **ABSENT** — the brief states the 24-port fabric as one pooled figure and does not separate intra-node from inter-node |
| `coherent_domain_size` | **ABSENT** |

**GC-TRN2 · GC-TRN3 · GC-INF2.** Source: the manufacturer's accelerator architecture documentation and EC2 instance pages · retrieved 2026-09-02 · quotation

| | Trainium2 | Trainium3 | Inferentia2 |
|---|---|---|---|
| `memory_capacity` | 96 GiB | 144 GiB | 32 GB |
| `memory_type` | HBM | HBM3e | HBM |
| `memory_bandwidth` | 2.9 TB/s | 4.9 TB/s | 820 GB/s |
| `coherent_domain_size` | 16, stated as pooled memory | **ABSENT** | **ABSENT** |
| `form_factor` · host interface · power | **ABSENT** | **ABSENT** | **ABSENT** |

**`link_bandwidth_per_gpu` is ABSENT for all three, and the reason is the same in each case: the manufacturer prints two different figures for it.** The chip documentation gives 1.28

TB/s per chip for Trainium2 and the instance table gives 1,024 GB/s per chip for what reads as the same quantity; Trainium3 is 2.56 TB/s per device in one document and 2 TB/s per chip in another; Inferentia2 is 384 GiB/s per chip in the architecture table and 192 GB/s on the product page. **None of the manufacturer documents cited reconciles any of the three pairs.** Both figures are recorded here and neither is entered.

This manufacturer prints per-chip specifications and sells instances. **The documents cited print no form factor, no host interface and no power figure for any of these parts**, which follows from the manufacturer selling them as instances rather than as cards a customer installs. Its pooling language is scoped to sixteen chips and is not extended to the larger server; whether that server is one pooled domain or four linked ones is left unstated.

## Annex 5 · Record schemas

The canonical form of every object this standard names is a published schema. **The schemas are those the reference implementation runs on; this standard does not define a second set.**

Canonicalization and signature follow the implementation: canonical JSON under the digest suite `psyche.digestsuite/1` — keys sorted, compact separators, ASCII-escaped — digested with SHA-256, and Ed25519 signatures.

| This standard | Schema |
|---|---|
| F1 Delivery Claim | `claim` — plus the I.1.4.1 pass identifier where the Profile states more than one pass |
| F2 Counterparty Decision | `claim_review` |
| Terminal state | `terminal_event` (`accepted` · `rejected` · `disputed` · `expired` · `withdrawn`) |
| Evaluation output | `sigma_output` (`claim-supported` · `claim-not-supported` · `evidence-fault` · `requires-review`) |
| C1–C5 Missing Intervals | `boundary_record` |
| B measurements, filed | `evidence_receipt` · `evidence_obligation` |
| A10 standard digest | `digest_suite` over the profile and its version, `profile_recipe` · `parameter_pack` · expected model · spot-check policy |
| A2 period function | `metric_registry_entry` — a 19086-2 Clause 8 metric plus the Annex 2.1.2 fields |
| A6 thresholds | `parameter_pack` |
| E2 inclusion proof | `settlement_head`, the fabric's signed head over a Case's settlement log · `oracle_log_head`, the head of Sphinx's spot-check issuance log |
| F3 authority reference | `grant`, the grant F2 was signed under, named by `claim_review.grant_id`; `authorization_snapshot` is the payer's authorization the release of money is checked against (§7.7 of the paper), not the review authority |
| Capsule and replay | `capsule`, with the verification material it is replayed from |
| Registry | `registry_snapshot` · common types |

A record that validates against these schemas is not thereby conforming. Conformance is Section 4. **A record that does not validate against them is not conforming under any reading.**

## Annex 6 · The evidence behind V1 and V2

A8.10 places some fields at V1 on the strength of a measurement the customer takes itself, and others at V2 on the strength of a signature a manufacturer stands behind. This annex states what each rests on, what each declines to carry, and what a Profile relying on either

must record.

**6.0 · The roles, and why they are borrowed rather than invented.** Attestation has a standard vocabulary and this standard uses it [RATS]. An **Attester** produces Evidence about itself. A **Verifier** appraises that Evidence against Reference Values under an appraisal policy and produces an Attestation Result. A **Relying Party** consumes the Result and acts on it. An **Endorser** — in practice the manufacturer — vouches for the Attester's capability, and a **Reference Value Provider** publishes the values the Verifier compares against.

Mapped onto this standard: for a V2 field the Attester is the device, the Endorser and the Reference Value Provider are the manufacturer, the Verifier is whatever appraises the report, and the evaluation is the Relying Party. **For a V1 field the customer is Attester and Verifier at once** — which is why Annex 7 must state the appraisal policy explicitly. 7.0.1's declared conditions and 7.0.2's bound are that policy, written out.

**6.0.1 · Where the borrowed model runs out.** That architecture assumes one Attester per fact. **This standard has two Filing Parties producing evidence about the same fact and disagreeing.** Section H exists for that situation, and the attestation architecture does not model it. The vocabulary is borrowed for the roles; the divergence machinery is this standard's own.

**6.0.2 · Two topologies, and why the choice is filed.** A Verifier may return its Result to the Attester, which presents it onward (the passport model), or it may sit behind the Relying Party and be consulted when the record is appraised (the background-check model). **The first survives the loss of the verification service. The second does not.** K1 requires a record to remain checkable for its retention period, and a Profile that depends on a vendor-hosted verifier at read time has made its record depend on that service still answering years later. **The Filing Plan MUST state which topology produced a V2 field**, and a record produced under the second SHOULD carry the Attestation Result itself rather than a reference to it.

**6.1 · The V1 primitive. The source.** [ATC], submitted 2026-06-22. A CUDA probe measures a streaming-multiprocessor-by-memory-region latency matrix using physical SM labels and dependent global loads. A reducer commits sufficient statistics, configuration, code hashes, network evidence and a compressed raw archive into a certificate **a verifier checks without a GPU**. Retrieved 2026-09-02.

**What it reports, and the device each figure was taken on.**

| Claim | Reported | Device |
|---|---|---|
| Fingerprint stability | median temporal jitter 0.09 cycles over a six-hour full-load run | **RTX 5090** |
| Die separation | 100.0 %, shape-normalised leave-one-out nearest-neighbour classifier | Blackwell dies |
| Topology-class recovery | unified memory domain · two-way L2 split · two-die package at a 30-cycle cross-die penalty | V100 · H200 · B200 |
| Location | within 44 km of the claimed datacentre, on 168 of 169 responding probes after the consensus fit | B200 |
| Location | within 270 km of the declared site, on 167 landmarks | V100 |

The stability figure was taken on a consumer part; the topology and location figures were taken on datacentre parts. **A standard that quotes 0.09 cycles against an H200 quotes a number nobody has measured on an H200.**

**The decoy result, at both of its stages.** Of eleven named decoy datacentre locations, round-

trip time alone rejects **nine**. All eleven are rejected after the consensus fit, which discards a probe reporting a physically impossible sub-millisecond round trip from a distant registered location.

A location test stopped at the first stage admits two false sites in eleven. MP-07 requires the second stage, and 7.1.6 states it.

A primitive that can locate is a convenience; a primitive that can **refuse a false claim** is evidence.

**What the source declines to claim.** The paper states that the certificate proves consistency of the measured evidence, and that it does not prove firmware integrity, prevent refusal to run, or replace vendor attestation where a trusted enclave is available. A host with physical control can cut power, refuse to run the probe, or avoid reachable services.

Refusal is a case this standard already holds. A probe the provider will not let run produces **no reading** under 7.0.3, enters the record as a BoundaryRecord under Section C, and is allocated under J1 — the treatment any other absent evidence receives. **V1 means that misreporting requires defeating a physical measurement.** Preventing the measurement is a different act, and it resolves as an evidence fault rather than as a reading.

**Independent precedent.** The technique has a peer-reviewed lineage this standard did not draw on and should name. Published work on remote GPU fingerprinting separates physically identical units of the same model by stalling execution units and measuring the resulting timing skew — unprivileged, in a browser, across more than two thousand devices and sixteen hundred distinct GPUs [DA]. **It corroborates the physics rather than the datacentre result**; the parts, the conditions and the purpose all differ. It establishes that individual-die timing variance is a real and exploited signal rather than a claim resting on one unreplicated paper.

**Status.** The construction has **not** been reproduced by this standard's maintainer. Until it is, the three fields stand at V1 **on an existence result** rather than on a deployed evidence class, and a Profile relying on them MUST record that under G8.

**Boundary.** The primitive establishes identity, class and coarse location. It reaches no further: exclusivity, contiguity across accelerators, bandwidth floors, firmware integrity and any quantity of delivered hours stay where A8.10 puts them.

**What remains owed.** Two things remain owed: independent reproduction on datacentre parts, the stability figure included, since that one was measured on a consumer card; and a false-accept rate for the landmark test. The source reports none, and reports one instance in which a second site remained acceptable alongside the correct one until colocation and autonomous-system evidence resolved it. Until a rate exists, MP-07 yields a metropolitan bound and a decoy rejection, and it carries no stated error.

## Annex 7 · Measurement procedures

A8.10 places a field at **V1** when the customer can establish its value by physical measurement. That is a statement about what is possible. This annex states what is required, because a value that two parties measure differently is not a measurement.

### 7.0 · What makes a reading admissible.

**7.0.1 · Declared conditions.** A reading carries the conditions under which it was taken: the

load state of the subject, whether the measurement window was exclusive, the driver and runtime versions in force, and the wall-clock start. **A reading whose conditions are not declared is not admissible**, whatever its value.

**7.0.2 · Uncertainty.** A reading is the median of $n$ runs, and it is reported with the uncertainty of that median rather than as a bare number.

Let $s$ be the median absolute deviation of the runs scaled by 1.4826, and let $u = 1.253 \cdot s \,/\, \sqrt{n}$ be the standard uncertainty of the reported median. The **expanded uncertainty** is $U = k \cdot u$, where $k$ is the coverage factor the procedure names. **A reading is admissible only where $U$ divided by the median falls inside the bound the procedure names, at the coverage factor the procedure states.**

The scaled median absolute deviation is a robust estimator of the standard deviation, and the factor 1.253 relates the uncertainty of a median to that of a mean for runs consistent with a single underlying distribution. A procedure MAY name a different estimator; it MUST name the one it uses.

**A single number with no uncertainty is not a measurement. It is an assertion with a unit attached.** Here this standard imposes a numeric requirement without the Profile's choice: an instrument's uncertainty is a property of the instrument, and allocates nothing between the parties. **What that uncertainty then decides is a different question, and 7.0.2.1 hands it back to the Profile.**

**7.0.2.1 · Conformity near a limit.** Where a reading is compared to a Grade Class floor or to the A5 threshold, the comparison uses the form the Profile selects from **Annex 2.8**. **SQCD-1 sets no default.**

**7.0.2.2 · Two parties, one field.** 7.0.2 bounds the agreement of a reading with itself. It says nothing about the agreement of two Filing Parties' independent readings of the same field. Section H is about that quantity.

Where both parties read the same V1 field, the Profile MUST state the **reproducibility limit** — the largest difference between two independent readings, taken by different parties on different instruments, to be treated as measurement rather than as divergence. **A materiality threshold under A6 applied to measured fields without a reproducibility limit turns instrument scatter into a Divergence Event**, and the parties then litigate their own noise.

**7.0.2.3 · Status of the bounds in 7.1.** The bounds this standard names in 7.1 are stated as relative expanded uncertainty at $k = 2$. **They rest on no interlaboratory study.** A reproducibility trial across independent operators and instruments would make them defensible, and until one exists a Profile that requires a tighter bound than 7.1 names is conforming and a Profile that requires a looser one MUST record the departure under G8.

**7.0.3 · Failure.** A run that cannot complete, or a set whose expanded uncertainty exceeds the bound, produces **no reading**. It does not produce a low reading. A failed measurement enters the record as a BoundaryRecord under Section C with cause `instrument-failure`, and never as evidence that the field was below its floor.

**7.0.4 · Cost.** A measurement consumes the subject. Each procedure states its duration and whether it requires exclusivity. Whether a measurement window counts against the availability obligation is a term of the agreement and MUST be fixed under A4.

**7.0.5 · Filing.** A reading is filed as an evidence receipt naming the procedure identifier, the conditions under 7.0.1, the $n$ runs, the median, the scaled median absolute deviation,

the coverage factor $k$, the expanded uncertainty $U$, and the Grade Class field it is offered against.

**7.0.6 · Traceability.** Each procedure in 7.1 states what its readings are traceable to.

Metrological traceability is the property of a result that relates it to a reference through a documented unbroken chain of calibrations, each contributing to the uncertainty [VIM]. **The procedures in 7.1 that measure a rate — MP-01 through MP-05 — have no such reference.** Their instrument is the subject under test or another node beside it, and no calibrated standard stands behind either. They establish **repeatability**: agreement of a reading with itself, under the conditions 7.0.1 requires it to declare.

A reading filed under MP-01 is evidence that a measurement was taken, was taken under declared conditions, and was reproduced within a stated bound. **It is not evidence that the quantity measured is what a calibrated instrument would have read**, because no such instrument exists for these quantities today.

**7.0.7 · Commencement and cadence.** A reading taken at commencement under B8.5 and a reading taken at cadence answer different questions, and the Profile MAY hold them to different bounds.

Legal metrology keeps this distinction: an instrument entering service is held to a tighter tolerance than the same instrument in service, so that a limited amount of drift is permitted before the instrument is rejected. **The commencement reading establishes that the Grade Class contracted for was the Grade Class delivered, and it is the baseline every later reading is read against.** Where the Profile states one bound for both, it MUST say so.

**7.1 · Procedures.**

| ID | Establishes | Probe | Runs | Bound, as $U$/median at $k = 2$ | Exclusivity |
|---|---|---|---|---|---|
| **MP-01** | `memory_bandwidth` | Sustained read and read-write sweeps over a working set exceeding last-level cache by ≥ 8× | ≥ 5 | 2 % | Per subject |
| **MP-02** | `link_bandwidth_per_gpu`, `coherent_domain_size` | Pairwise peer-to-peer bandwidth matrix across all accelerators on the subject | ≥ 3 full matrices | 3 % per ordered pair | Per subject |
| **MP-03** | `fabric_bandwidth_per_gpu` | Pairwise inter-node transfer across the allocation, all-to-all and ring | ≥ 3 | 5 % | Across ≥ 2 subjects held simultaneously |
| **MP-04** | `storage_throughput_floor`, `storage_iops_floor` | Sequential and random access at a queue depth the procedure names, working set exceeding any cache tier | ≥ 5 | 5 % | Per subject |
| **MP-05** | `egress_floor`, `ingress_floor` | Sustained transfer against an endpoint outside the provider's network, named in the Filing Plan | ≥ 5 | 10 % | None |
| **MP-06** | accelerator identity, `die_topology` class | Per-streaming-multiprocessor latency map and cache-bypassing memory sweep, per Annex 6 | ≥ 3 | Identity: match to the 7.1.5 baseline. Class: unanimous across runs | Per subject |
| **MP-07** | coarse geographic position | Round-trip timing against public network landmarks, with decoy sites named in advance | ≥ 1 landmark set | Landmark set stated | None |

**7.1.1** `coherent_domain_size` is read from MP-02 as the largest set of accelerators mutually reachable at the link bandwidth of A8.2, rather than at host-bus bandwidth. **A domain is**

**what the instrument finds, not what the control plane reports.**

**7.1.2** MP-03 requires the customer to hold two subjects at once. A customer holding one subject cannot establish fabric bandwidth, and the field is V4 for that customer for the duration.

**7.1.3** MP-07 is admissible only where the landmark set and the decoy set were **named before the reading was taken**. A location claim tested only against landmarks chosen afterwards establishes nothing, since the choice can follow the answer.

**7.1.4** A procedure establishes that a field met its floor. **It does not establish that the field held between readings.** Continuity is a matter for cadence under B8.5, and a gap between readings is a gap.

**7.1.5 · MP-06 baseline.** An identity reading is a comparison, so it requires something to compare against. The baseline is the per-subject fingerprint taken at commencement under B8.5 and committed under the A10 digest before the Window opens. **An identity reading taken against a baseline committed after the Window opened establishes nothing**, since a baseline recorded later can follow the substitution it would otherwise reveal. The `die_topology` class is established where three runs return one class and no run returns another.

**7.1.6 · MP-07 consensus.** A location reading is admissible only where the fit discards probes whose reported position is inconsistent with their own round-trip time, and the decoy set is scored on the probes that survive. In the source's experiment, round-trip time taken alone left two of eleven decoys standing (Annex 6).

**7.2 · What the procedures do not reach.** `allocation_contiguity`, `tenancy`, `preemptibility` and rack-level position have no procedure here. They are V4 under A8.10 and carry a required evidence class under A8.11.

MP-02 finds which accelerators are mutually reachable at link bandwidth. **It does not find where they are.** MP-07 places a subject within a metropolitan bound. **It does not place it in a rack.** No procedure in this annex reports another tenancy.

This annex is the boundary of what measurement does for a delivery record under this standard. Everything beyond it is a signature or nothing.

## Annex 8 · Profile conformance checklist

Every term this standard requires a Contract Profile to fix, with the clause that requires it and the set it is drawn from. **A Profile missing any row marked required cannot be armed**, and the omission is detectable when it is armed rather than when it is disputed.

Rows 25 to 29 are required by this annex itself, and for them the Clause column names the section each term attaches to. Every other row restates a requirement of the clause it cites.

### 8.1 · Required — the Profile cannot be armed without these.

| # | Term | Clause | Drawn from |
|---|---|---|---|
| 1 | Obligation reference | A1 | the agreement |
| 2 | Period function | A2 | metric registry, `canonical` or declared `experimental` |
| 3 | Measurement window, offset, edge treatment | A3 | the agreement |
| 4 | Exclusions, each with the evidence required to invoke it | A4 | the agreement |
| 5 | Tolerance and threshold | A5 | the agreement |
| 6 | **Materiality threshold** — form and parameter | A6 | **Annex 2.2** |

| # | Term | Clause | Drawn from |
|---|---|---|---|
| 7 | **Unresolved ceiling** — form and parameter | A6 | **Annex 2.3** |
| 8 | Remedy schedule reference | A7 | held where the evaluation cannot read it |
| 9 | **Grade Class** — every field valued or marked `not-material` | A8 | Annex 4, or declared |
| 10 | **Required evidence class for every V4 field** | **A8.11** | the agreement |
| 11 | Substitution factor, where any is permitted | A9.2 | **Annex 2.6** |
| 12 | **Standard digest, committed before the Window** | **A10** | computed over A1–A9 |
| 13 | Measured Subject grain | B3 | the agreement |
| 14 | **Covered Layers** | B7.1 | L1–L7; L8 is never covered |
| 15 | Minimum customer-side observation | B8.2 | the agreement, or B8.2's default |
| 16 | V1 reading cadence | B8.5 | the agreement |
| 17 | **What expiry costs the silent party** | **I.1.1** | **Annex 2.4** |
| 18 | **Evidence-fault allocation, per obligation** | **J1** | **Annex 2.5** |
| 19 | **Divergence resolution** — form and its parameters | **H4** | **Annex 2.7** |
| 20 | **Conformity decision near a limit**, per measured field | **7.0.2.1** | **Annex 2.8** |
| 21 | **Reproducibility limit**, where both parties read one V1 field | **7.0.2.2** | the agreement |
| 22 | Attestation topology, per V2 field | 6.0.2 | passport or background-check |
| 23 | Review deadline | I.1 | the agreement |
| 24 | Recomputation passes, where more than one | I.1.4 | the agreement; one pass where unstated |
| 25 | Settlement policy quorum | F2 | order-insensitive forms only |
| 26 | Challenge policy and post-finality window | K | the agreement |
| 27 | Grant constitution | F3 | absence is refusal, not permission |
| 28 | Head publication frequency and gossip obligation | E2 | at minimum, one head per Window close |
| 29 | Value at risk for the obligation | F1 | the agreement |
| 30 | Whether a measurement window counts against availability | 7.0.4 | fixed under A4 |
| 31 | Retention period | K1 | not less than this standard's floor |
| 32 | Declared departures from this standard | G8 | **"none" is a permitted entry** |

**8.2 · The seven that decide who pays.** Rows 6, 7, 10, 17, 18, 19 and 20 are the ones this standard refuses to fill.

Each settles who bears a cost when the record is partial, silent, contested, or measured close to a limit. **The standard requires of those seven that they exist, that they are fixed before the Window opens, and that all except row 10 are selected from a published set.**

Row 18 additionally carries a merits test. Under 2.5.1 a selection is refused where the party holding the required evidence is the party that gains from its absence. It refuses only the case where a party is paid to stay blind.

**8.3 · Detection.** Rows 1–32 are checkable against a Profile without reference to any evidence, any Window, or any counterparty. **A Profile defect is therefore an arming-time finding, not a dispute-time discovery**, and a compiler that arms Profiles is the right place to enforce this annex.

The alternative is the ordinary case. A term nobody fixed becomes a term somebody argues, six weeks after the month closed, on a call.

## Annex 9 · Conformance vectors

Section 4 states what conformance means. The vectors this annex lists check it.

Published with this standard, under the permissive licence of 0.2:

| File | Contents |
|---|---|
| `annex3-worked-month.json` | Annex 3's month as a fixture — the standard in force, the five events, and every figure the record produces. A conforming implementation reproduces all of them from the event counts alone. |
| `edge-cases.json` | Fourteen cases stating an input and the only conforming outcome, each naming the clause it tests and what a conforming implementation MUST NOT return. |
| `MANIFEST.sha256` | A digest per file. |

**9.1** A vector states a required outcome and carries no authority beyond the clause it names. Where a vector and the clause it tests disagree, **the clause governs and the disagreement is an erratum under L6**.

**9.2** Passing these vectors makes an implementation consistent with the cases they cover. **It establishes nothing about a case they do not cover**, and this set covers the arithmetic of one worked month and fourteen boundary conditions.

**9.3** No party certifies an implementation under this standard, and PSYCHE operates no conformance programme. **A claim of conformance is the claimant's own**, made against a named version, and these vectors are what a counterparty can run against it without asking anyone's permission.

## References

Two kinds of entry. **Normative** references are those a conforming Profile or record depends on. **Informative** references support an argument this standard makes and bind nothing.

A reader who cannot obtain a reference cannot check the clause resting on it, so a freely available source is named wherever one carries the same substance. Where a reference is available only at cost and no free equivalent could be found, this standard says so rather than name a substitute that says something else.

### Normative.

| Tag | Reference |
|---|---|
| [**19086-1**] | ISO/IEC 19086-1:2016, *Cloud computing — Service level agreement (SLA) framework — Part 1: Overview and concepts*, with Amendment 1:2025. Free from ISO. |
| [**19086-2**] | ISO/IEC 19086-2:2018, *Part 2: Metric model*, with Amendments 1:2023 and 2:2025. Free from ISO. Clause 8 and its normative XML schema are the basis of Annex 2.1. |
| [**22123-1**] | ISO/IEC 22123-1:2023, *Cloud computing — Part 1: Vocabulary*. Free from ISO. Supersedes ISO/IEC 17788:2014. The source of *cloud service provider* and *cloud service customer* as used here. |

### Informative.

| Tag | Reference |
|---|---|
| [**HVR**] | The Hague Rules as amended by the Brussels Protocol 1968 (the Hague-Visby Rules), Article III. Public international convention; the text is scheduled to national carriage-of-goods statutes and is freely available. Cited in Section 2. |
| [**UCP600**] | ICC Uniform Customs and Practice for Documentary Credits, 2007 revision, Article 34. Published by the ICC at cost. Article 34 is reproduced verbatim in *Abani Trading Pte Ltd v BNP Paribas* [2014] SGHC 111, free from the Singapore judiciary. |
| [**CSCC**] | Cloud Standards Customer Council / OMG Cloud Working Group, *Cloud Service Agreements: What to Expect and What to Negotiate*, Version 3.0, 2019, p. 20. Free from omg.org. Cited in Section 2 for current practice on service-credit claims. |
| [**BJPD**] | *Brown Jenkinson & Co Ltd v Percy Dalton (London) Ltd* [1957] 2 QB 621 (CA). Cited in Section 2 for the effect of a knowingly false clean bill issued against a letter of indemnity. |

| Tag | Reference |
|---|---|
| **[GUM]** | JCGM 100:2008, *Evaluation of measurement data — Guide to the expression of uncertainty in measurement*. Free from the BIPM. |
| **[VIM]** | JCGM 200:2012, *International vocabulary of metrology*. Free from the BIPM. The source of *metrological traceability* as used in Annex 7. |
| **[JCGM106]** | JCGM 106:2012, *The role of measurement uncertainty in conformity assessment*. Free from the BIPM. |
| **[RATS]** | IETF RFC 9334, *Remote ATtestation procedureS (RATS) Architecture*. Free from the IETF. The source of the attestation roles named in Annex 6. |
| **[ATC]** | F. Alpay and T. Alpay, *Unprivileged Topology Certificates for Cloud GPU Attestation*, arXiv:2606.24934 [cs.CR], 2026. Preprint. Cited in Annex 6 as an existence result. |
| **[DA]** | T. Laor et al., *DrawnApart: A Device Identification Technique based on Remote GPU Fingerprinting*, NDSS 2022. Cited in Annex 6 as independent precedent for the measurement technique. |

**Named, not relied on. ISO/IEC 19086-3:2017**, *Part 3: Core conformance requirements*, is the part of that family nearest to Section 4 and Annex 8 of this standard. It is available only at cost, and no free source stating its requirements could be located as of 2026-09-03. **Section 4 was written without it**, and this standard makes no claim about the relationship between the two.

Send a closed case and PSYCHE will reconstruct it under SQCD-1.